

Mayoogh Girish

Product Development Lead - Hardware

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PROFESSIONAL SUMMARY

PCB layout designer and embedded hardware engineer with 5+ years of experience designing compact, multilayer PCBs for industrial AGV robotics and consumer electronics. Hands-on ownership of the complete PCB design process flow. Currently leading the Electronics Department at Flexli Technologies, with production systems deployed at Sitics Logistics warehouse supporting Meesho parcel operations.

WORK EXPERIENCE

FLEXLI TECHNOLOGIES PVT LTD

Embedded Development Engineer.

Bangalore, KA

July 2022 - Present

Project : E-commerce warehouse package sortation robots

- Leading the embedded hardware and electronics development activities for autonomous mobile sortation robots deployed at Sitics warehouse operations handling Meesho parcels.
- Responsible for the overall electronics department, including hardware development, PCB design, validation, production coordination, and deployment support.
- Designed and developed a compact 4-layer motherboard for AGV-based sortation robots with optimized component placement and space-efficient PCB layout.
- Reduced motherboard size by 60% and lowered production cost by 45% through PCB optimization and efficient component selection.
- Designed compact sensor PCBs under strict mechanical size constraints for robotic applications.
- Worked extensively on schematic review, ERC/DRC validation, PCB layout optimization, routing, and board bring-up activities.
- Performed unit testing, hardware debugging, and validation during prototype and production stages.
- Familiar with PCB manufacturing release workflows including Gerber generation, BOM validation, component sourcing, and order release through vendor platforms such as JLCPCB.
- Designed and developed custom cable harness systems for AGVs to improve assembly, maintenance, and troubleshooting efficiency.
- Collaborated closely with cross-functional Mechanical and Software teams during product development, integration, and deployment.
- Managed electronics production activities to ensure product quality, manufacturability, and timely delivery.

OTOMATOR TECHNOLOGIES

Embedded Developer

Remote (Noida)

April 2021 - 2022

Project : Home automation product development

- Worked on end-to-end electronic product development for home automation systems.
- Designed compact PCB layouts and schematics for products intended to fit inside standard wall switch boards with strict space constraints.
- Focused on compact enclosure and PCB co-design to achieve manufacturable and space-efficient product integration.
- Performed schematic reviews, PCB debugging, and hardware validation to maintain product quality and reliability.
- Conducted component selection activities considering performance, size, cost, and availability requirements.
- Worked on enclosure development and rapid prototyping using FDM 3D printing and vacuum casting methods.
- Gained hands-on exposure to product manufacturing workflows and prototype enclosure fabrication processes.
- Collaborated across hardware and product development activities from concept to prototype realization.

EDUCATION

Christ University Faculty of Engineering
Bachelor Of Technology In Electronics and Communication
Cumulative GPA: 2.78/4.0

Bangalore, KA
2016 – 2022

PROJECTS

UV Based Upper Air Disinfection System (Freelance)

June 2021

Developed an ozone-free UV-C LED-based Upper Air Disinfection System (UADS) for HVAC and indoor environments, achieving 99.9% disinfection efficiency for air purification and pathogen control.

- Designed the PCB for the control board used in the UV-Based Upper Air Disinfection System.
- Developed custom UV-C LED strips using aluminum-core PCBs for efficient thermal management and compact integration.
- Worked on embedded firmware development for system control and operation.
- Contributed to hardware development, PCB validation, and testing activities for the ozone-free UV-C air disinfection system.
- Supported development of a high-efficiency air purification solution for HVAC and indoor environments

Split Parallel Hybrid Electric Vehicle (Christ University Faculty of Engineering)

Jan 2020

The "Split Parallel Hybrid Electric Vehicle" project involves developing a working prototype of a hybrid vehicle that combines an internal combustion engine (ICE) and an electric motor. The system is designed to analyze the route and terrain using geo-mapping technology to implement an automatic switching algorithm. This algorithm dynamically switches between the ICE and the electric motor to achieve optimal efficiency and performance. The project aims to demonstrate the benefits of split parallel hybrid technology, ensuring efficient fuel usage and reduced emissions while providing a smooth and responsive driving experience.

3D printed prosthetic hand using EMG signals (Personal)

Jan 2022

The "3D Printable Prosthetics" project focuses on creating customizable and adaptable prosthetic limbs using 3D printing technology. Traditional prostheses are custom-made for each individual, often leading to high costs and difficulties in replacement, particularly for children who outgrow their prosthetics. This project addresses these challenges by enabling easy fabrication and resizing. The system employs electromyography (EMG) signals to capture and learn muscle patterns during different activities. These signals are then amplified and analyzed by a microcontroller, which controls servos to manage the movement of the prosthetic fingers. This approach not only reduces costs but also allows for rapid adjustments and custom designs, making prosthetics more accessible and adaptable to the user's needs.

[Controlling a 3D printed prosthetic hand using EMG signals](#) (Video)

Automated Bell Scheduler (Christ University Faculty of Engineering)

Jan 2017

Designed and implemented an automated bell scheduler for a university. The new system was developed to streamline scheduling and operations, resulting in approximately a 50% reduction in cost compared to the existing system. This improvement not only enhances efficiency but also significantly reduces operational expenses.

TECHNICAL SKILLS

- **eCAD Tools:** KiCAD (primary), Altium Designer (familiar), PSpice (familiar), Autodesk Fusion 360
- **PCB Design:** Schematic capture, 4-layer multilayer PCB layout, Component placement & routing, ERC/DRC validation, DFM/DFA constraint review, Gerber generation & release, BOM validation & vendor portal sourcing (JLCPCB), Layer stack-up & size optimisation, Board bring-up & unit testing
- **Interfaces:** I²C, SPI, CAN, UART
- **MCU / SoC:** AVR, ESP8266, ESP32 (Espressif)
- **Languages:** C/C++, Python
- **Manufacturing:** FDM 3D Printing, Vacuum Casting
- **Version Control:** Git

AWARDS & RECOGNITION

- KPIT Sparkle Finalist Top 30 (2020) (National Level)
 - First Prize, IEC Tech Fest, Christ University, 2017 (University)
 - Third Prize, IEC Tech Fest, Christ University, 2020 (University)
 - Third Prize “Young Engineer” (2015) NIT Calicut
 - Third Position in Jawaharlal Nehru National Science Exhibition (2014)
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